

# LDaCA Program Persistent ID Policy, Strategy and Process

Updated: 2024-06

## Purpose

This document outlines the policy, strategy and process on the use of persistent identifiers for data in the Language Data Commons of Australia project (LDaCA).

## Scope

This policy covers data collections from participating organisations that will be made available on the LDaCA portal and all research output such as software and computational notebooks published by members of the LDaCA program and ATAP projects until the end of the project.

Comments and suggestions received during meetings among ARDC and LDaCA representatives, documented in the internal document *LDaCA Program Persistent Identifiers (PIDS) for Data: Strategy* have been taken into consideration.

## Definitions

**Data Steward:** An individual or organisation with the authority to make decisions regarding the Collection.

**Content host/custodian:** An individual or organisation who holds the content with or without delegated authority to make decisions about the content. Probably the only decisions they can make are of a technical nature within ethical & legal constraints, e.g. move content to a new location.

**Custodial Organisation:** This may be an institution that hosted or is affiliated with the research project that developed the language data collection, an organisation with an interest in the research, a central repository, or a repository selected by the data steward.

**Participating Organisation:** An organisation or institution that agrees to make their data available on LDaCA.

# Policy

LDaCA and ATAP support the recommendations outlined in the *ARDC Persistent Identifiers Policy* in the use of persistent identifiers (PIDs).

Consistent with the aim of LDaCA and ATAP to assist their participating institutions in building repository infrastructure, LDaCA will follow a participating institution's existing PID practice to ensure DOIs or other identifiers can be used to find resources within the LDaCA repository, using the persistent ID as a locally resolvable identifier so that there is less reliance on global resolution services.

In the absence of existing policy, LDaCA repositories will encourage custodial organisations to use DOIs to identify collection data and maintain landing pages for records that point to LDaCA and any other data holdings that are held in LDaCA repositories/portals. This PID strategy, which is outlined below, is designed so that:

- Citations of DOIs are concentrated at a level that will help demonstrate the impact of particular collections or objects; that is, there will be numerous citations of a few PIDs rather than a few citations of each of numerous PIDs.
- Requirements to update DOI URLs are not too onerous for data managers over the long term.

## Strategy and Process

LDaCA will request participating institutions to provide a collection dataset PID and information about its existing PID practice. If there is no PID practice in place, LDaCA will recommend using DOIs and provide guidance on how to get a DOI for the collection data, as well as governance considerations for a sustainable PID system.

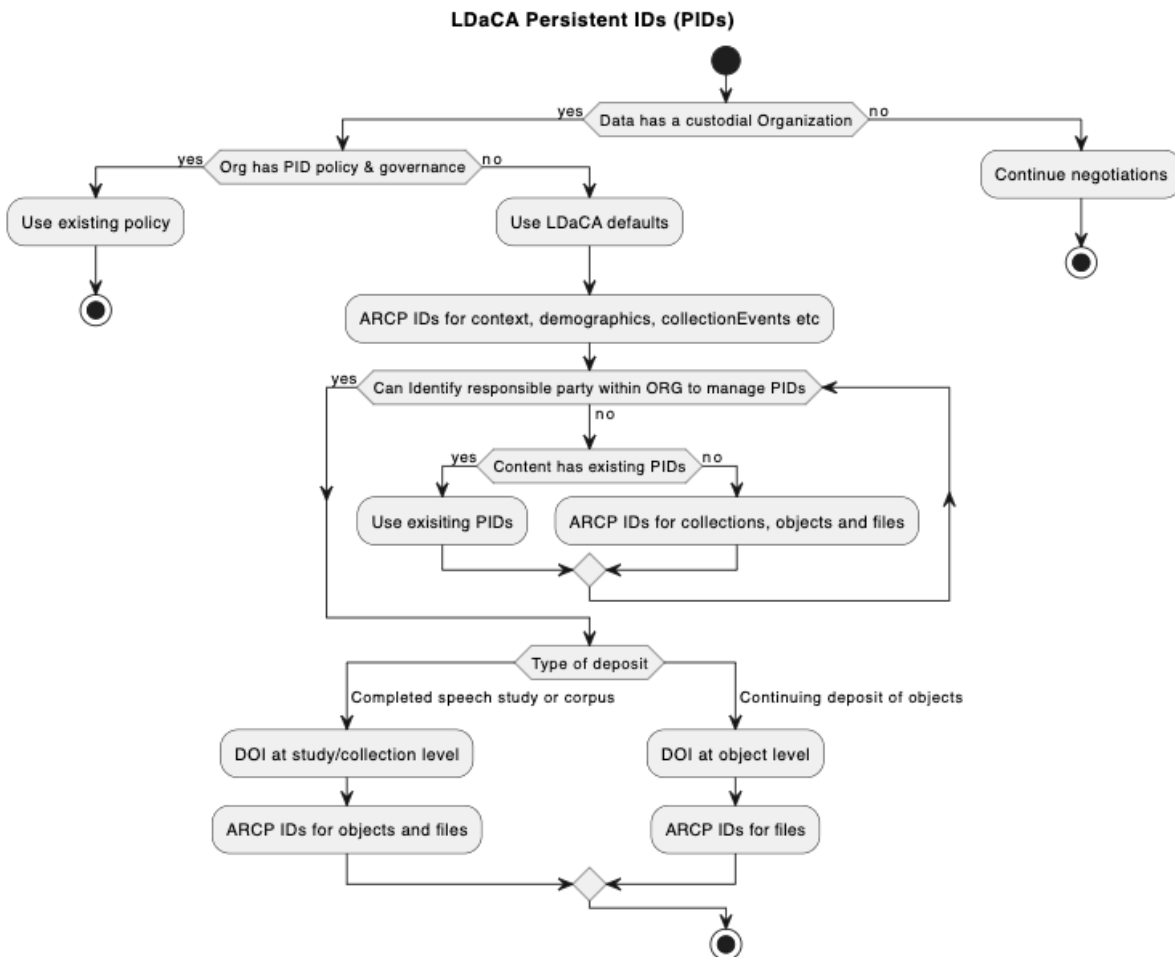
Steward and Custodian roles in participating organisations must be defined and a custodial organisation identified, as these entities will be responsible for the management and maintenance of PIDs for the life of the dataset. For example, a DOI must resolve to a publicly available landing page that contains metadata about the collection dataset and a link to access the data – maintenance of the DOI landing page is part of the long-term management of the collection's PID system.

LDaCA can begin establishing repository infrastructure while an organisation does not have a PID policy in place, by using Archive and Packaging IDs (ARCP) which are globally unique, searchable and have zero management overhead, but which do not resolve to content in a browser; hence, LDaCA recommends that a dataset have at least a DOI for the collection.

Recognising that identifier assignment practice will inevitably differ between organisations, LDaCA will design systems to make it as simple as possible to cite data - with clear "Cite as"

advice on data portals and in code such as notebooks. LDaCA and ATAP consider training users in citation practice as part of this process.

## LDaCA Persistent Identifier Strategy



The above strategy may have drawbacks that work against the concept of reproducibility in research culture, for example:

- Large datasets may change over time as errata are fixed, data are withdrawn or added etc.
- A high-level DOI does not capture which versions of which resources were actually used.

To overcome these drawbacks, LDaCA and ATAP will build systems and training that encourage researchers and make it straightforward to:

1. Take responsibility for creating their own more detailed record of the data they use in the form of an open-access “data publication”. This may take several different forms, depending on the circumstances of use:
  - a notebook or other code artefact that contains an embedded data package with links to fetch external data from LDaCA, ATAP or other data stores (e.g. Zenodo) with explicit versioning where possible;
  - a stand-alone data package with explicit links to resources available on an API;
  - or, for data sources which are not available in a stable form from a trusted repository (such as being aimed to build in LDaCA and ATAP), resources (files) themselves should be packaged with metadata. An example of this would be Trove resources which are not available on a version-controlled API.
2. Deposit the open data publication in places that can provide a PID such as:
  - a public repository (e.g. Zenodo);
  - a discipline repository (e.g. an LDaCA repository (should this use case gain sufficient support) or PARADISEC);
  - an institutional data repository.

As part of the role of the LDaCA Program and ATAP in training users in citation practice, researchers will be encouraged to take responsibility by providing both education and services that allow them to produce data publications that list the datasets they used (using their PIDs), and reference the parts of those resources.

This kind of data publication could take many forms, but initially, we are working on two approaches:

1. A Jupyter notebook based Data Publication, where a git repository containing a notebook can be annotated with metadata, and published to a repository as an RO-Crate that cites its data sources using PIDs (where available), and specific API calls to get the relevant parts of the sources, such as files. Tools to support this are under active development as of August 2022.
2. A more general RO-Crate package that is a wrapper for a set of data which may be included in-crate if size and licensing constraints permit, or referenced by URL if not. (This is possible at the moment to compile by hand but we do not have tool support).

# Roles/Responsibilities

|                            |                                                                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDaCA Program / ATAP       | Implement a PID strategy that aligns with ARDC recommendations while taking into consideration established PID practices of participating organisations.              |
| Participating organisation | Maintain PIDs of own collection/data objects, manage related persistence instruments and governance arrangements for the life of the collection/data objects.         |
| Custodial organisation     | Academic institution or an organisation that funded, sponsored, published, approved or had/has stakeholder interest in the research activity that collected the data. |

# Document Metadata

|                    |  |
|--------------------|--|
| Web Links          |  |
| Approval Authority |  |
| Date Completed     |  |
| Next Review Date   |  |
| Audience / Users   |  |
| Notes              |  |

# Appendix

## Notes on DOI resolution and referencing parts of objects

FAIR signposting: LDaCA will work with the RO-Crate team to implement “FAIR signposting” which is a way for repository landing pages for DOIs to automatically indicate where data may actually be found; implementing advice in the forthcoming RO-Crate 1.2 spec

<https://github.com/ResearchObject/ro-crate/issues/160>

We are also exploring using ARCP IDs to provide links to the contents of DOIs (which is not currently possible). Such an ID might look like this:

```
arcp://doi,10.5281 zenodo.1312582/some/part/of/the/data
```

The URI is not resolvable in a browser, but via FAIR signposting, an algorithm could

- Extract the DOI <https://10.5281/zenodo.1312582>
- Resolve the DOI to a service such and via an as-yet not invented mechanism or convention, be able to resolve the ARCP  
<http://data.atap.edu.au/?id=arcp://doi,10.5281 zenodo.1312582/some/part/of/the/data>